

Table 7: Example prompts for generating story with WritingPrompt Premise.

Prompt Type	Example
How we generate premise based on two stories	Write a continuous, immersive literary fiction story of at least {length} words. The story should unfold in a natural, human-like way--flowing as if written by a thoughtful novelist, not an AI. Premise: {prompt} The story should: Use natural pacing: weave moments of quiet reflection with vivid external events, instead of nonstop introspection. Balance internal monologue with dialogue and interaction, so the protagonist feels alive and connected to others. Maintain a consistent atmospheric tone with sensory detail (sounds, smells, textures) that feel authentic rather than overly ornamental. Prioritize human readability: avoid mechanical repetition, vary sentence lengths, and use subtle rhythm to make the prose engaging. Escalate tension gradually, but allow for moments of relief, small human connections, or fragile beauty. Important: Write as if you are crafting a novel that would be highly rated by human readers. The prose should feel alive, compassionate, and deeply human, not robotic or overly stylized.

Table 8: Kurtosis of story chosen by investigated models, STORYREWARD and STORYRMB. STORYREWARD-QWEN (on wrong predictions) represents those stories different with stories chosen by STORYRMB. STORYREWARD-QWEN (on correct predictions) represents those stories being the same as stories chosen by STORYRMB. "Difference" means relative difference of kurtosis between stories chosen by other models and stories chosen by STORYRMB.

Model	Kurtosis	Difference (%)
STORYREWARD-QWEN (on wrong predictions)	1.0149	+12.8
STORYREWARD-QWEN (all)	1.0064	+11.8
STORYREWARD-QWEN (on correct predictions)	0.9958	+10.7
GRM-Llama3.1-8B-rewardmodel-ft	0.9573	+6.4
Skywork-Reward-Llama-3.1-8B-v0.2	0.9220	+2.4
QRM-Llama3.1-8B-v2	0.9056	+0.6
STORYRMB (ground truth chosen story)	0.8999	0.0
Skywork-Reward-Gemma-2-27B-v0.2	0.7598	-15.6
ArmoRM-Llama3-8B-v0.1	0.6876	-23.6

Table 9: Dataset statistics for our training data and STORYRMB.

Dataset	#Instances	Source	Average Length (words)	Median Length (words)
Training data	93,729	Douban, WritingPrompts, LLMs	2,802	1,858
STORYRMB	1,133	WritingPrompts, LLMs	3,011	2,401

Table 10: Results on Creative Writing v3 test set of EQ-Bench (Paech, 2023) with a widely-used evaluation method for story generation (Chhun et al., 2024).

Model	Avg Score
STORYREWARD-QWEN	2.72
STORYREWARD-LLAMA	2.51
GRM-Llama3.1-8B-rewardmodel-ft	2.24
Skywork-Reward-Llama-3.1-8B-v0.2	2.24
QRM-Llama3.1-8B-v2	2.31
Skywork-Reward-Gemma-2-27B-v0.2	2.28
ArmoRM-Llama3-8B-v0.1	2.36